

02 — Simulation: families, prior, constraints

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1 Theory

Aim

- One pool of patterns, swept over the regime coordinates $(\bar{n}, \tilde{\delta})$; train and test are carved from it later
- Same $(\bar{n}, \tilde{\delta})$ distribution in every structured family $\Rightarrow$ the regime carries no information about the family
- Two patterns per $\theta \Rightarrow$ bias/variance at fixed θ
- $\tilde{\delta}$ as defined in 01 (studentised sup of $L - r$, uncentred, at $\bar{n}$)

Families

Each $\theta = \bar{n} + \text{shape}$ (drawn) + *amplitude* (solved for the target $\tilde{\delta}$).

Family	Model	Shape	Amplitude
Poisson	homogeneous, intensity $\bar{n}$	—	—
Thomas	parents κ ; Poisson(μ) offspring, $N(0, \sigma^2 I)$; $\bar{n} = \kappa\mu$	σ	μ
Nested	meta-parents κ ; Poisson(μ_1) parents, $N(0, \sigma_1^2 I)$; Poisson(μ_2) children, $N(0, \sigma_2^2 I)$; $\bar{n} = \kappa\mu_1\mu_2$	$\sigma_2, \rho = \sigma_1/\sigma_2, \mu_1$	μ_2
LGCP	$\Lambda = e^{\mu+Y}$, $\text{Cov } Y = \sigma^2 e^{-r/s}$; $\bar{n} = e^{\mu+\sigma^2/2}$	s	σ^2
Matérn II	primaries λ_p , hard core R (type II); $\bar{n} = (1 - e^{-\lambda_p \pi R^2})/(\pi R^2)$	—	R

Closed-form $K(r) - \pi r^2$

- Thomas: $(1 - e^{-r^2/4\sigma^2})/\kappa$
- Nested: $(1 - e^{-r^2/4\sigma_2^2})/(\kappa\mu_1) + (1 - e^{-r^2/4(\sigma_1^2+\sigma_2^2)})/\kappa$
- LGCP: $2\pi \sum_{k \geq 1} \frac{\sigma^{2k}}{k!} \frac{s^2}{k^2} [1 - e^{-kr/s}(1 + kr/s)]$ (80 terms), from $g = \exp(\sigma^2 e^{-r/s})$
- Matérn II: $g = 0$ below R , 1 beyond $2R$, Stoyan's closed form between; K by trapezoid on $[R, 2R]$
- Monotone in the amplitude at fixed $(\bar{n}, \text{shape})$ (checked on grids) $\Rightarrow$ amplitude for a target $\tilde{\delta}$ unique

Validity rules

Two numbers: $\varepsilon = 0.2$, $c = 0.45$. Radius set $R(\bar{n}) = [r_{\min}(\bar{n}), 0.25]$ as in 01.

1. **Scale inside** $R(\bar{n})$ (shape only). Structure correlation $\gamma(r) \in [0, 1]$ must satisfy $\gamma(r_{\min}(\bar{n})) \geq \varepsilon$ and $\gamma(0.25) \leq \varepsilon$
 - Gaussian ($\gamma = e^{-r^2/4\sigma^2}$): $\sigma \in [r_{\min}(\bar{n}), 0.25]/2\sqrt{\ln(1/\varepsilon)} \Rightarrow \sigma \leq 0.098$
 - Exponential ($\gamma = e^{-r/s}$): $s \in [r_{\min}(\bar{n}), 0.25]/\ln(1/\varepsilon) \Rightarrow s \leq 0.155$
 - Nested: inner σ_2 for the lower bound, outer $\sqrt{\sigma_1^2 + \sigma_2^2}$ for the upper bound
 - Why: $\tilde{\delta}$ only sees $R(\bar{n})$; scale is averaged over within regimes, fair only if it is visible everywhere

2. Nested levels distinguishable (shape only)

- Separation: inner γ at ε before outer γ at $1 - \varepsilon \Rightarrow \sqrt{1 + \rho^2} \geq \sqrt{\ln(1/\varepsilon)/(-\ln(1 - \varepsilon))}$, $\rho \geq 2.49$
- Balance: inner share of the K plateau $1/(1 + \mu_1) \geq \varepsilon \Rightarrow \mu_1 \leq 4$; $\mu_1 \geq 1$ (interpretability)

3. Enough independent units (involves the amplitude): $\text{CV}(n) = \text{sd}(n)/\bar{n} \leq c$

$$\text{CV}^2 = \frac{1}{\bar{n}} + \int_0^1 \bar{\gamma}_W(r) d(K(r) - \pi r^2), \quad \bar{\gamma}_W(r) = 1 - \frac{4r}{\pi} + \frac{r^2}{\pi}$$

- $\bar{\gamma}_W$: isotropised set covariogram of W ; formula vs simulated Thomas CV: 0.409/0.413, 0.310/0.306, 0.444/0.445
- Why: $\tilde{\delta}$ is evaluated at $\bar{n}$, meaningful only if $n \approx \bar{n}$
- Old DV3 constraints in these terms: Thomas 0.41–0.44, nested 0.26–0.31, LGCP up to 0.72 (inconsistent)

4. Matérn II packing: $\pi R^2 \bar{n} \leq 0.95$ ($\lambda_p \leq 3.15 \bar{n}$); hard limit is 1

With $\varepsilon = 0.2$ the old DV3 bounds are recovered: $\sigma \leq 0.1$, $s \leq 0.15$, $\rho \geq 3$. The LGCP lower bound widens (0.5 spacings $\rightarrow$ 0.16–0.055).

Feasibility and the $\tilde{\delta}$ range

Largest $\tilde{\delta}$ reachable under the rules (`scripts/feasibility.py`):

$\bar{n}$	100	152	230	348	528	800
Thomas	16.8	26.9	42.3	65.5	101	155
Nested	14.4	23.9	38.7	61.7	97.1	150
LGCP	17.3	27.4	42.3	63.9	95.6	148
Matérn II	4.3	5.3	6.5	8.0	9.9	11.9

- Cluster families $\propto \bar{n}$ (CV fixes the clustering, $s_0 \propto 1/n$); Matérn $\propto \sqrt{\bar{n}}$ (packing limit)
- Matérn cap physical: fill 0.95 $\rightarrow$ 0.999 moves it 4.34 $\rightarrow$ 4.45 at $\bar{n} = 100$
- Common range at every $\bar{n} \Rightarrow \tilde{\delta} \in [\delta_{lo}, 4]$; $\bar{n}$ and $\tilde{\delta}$ independent
- $\delta_{lo} = 0.1$ for now; any larger δ_{lo} is a subset of this pool (log-uniform restricts to log-uniform)

Prior (draw order is part of the definition)

1. $\bar{n} \sim \text{log-uniform}[100, 800]$
 2. $\tilde{\delta} \sim \text{log-uniform}[0.1, 4]$ (Poisson: $\tilde{\delta} = 0$, skip 2–4)
 3. Shape: each coordinate log-uniform on its rule-1/2 box at $\bar{n}$; redraw until valid, $\tilde{\delta}$ reachable, and (LGCP) a grid exists
 4. Amplitude: root of $\tilde{\delta}(\bar{n}, \text{shape}, a) = \text{target}$, in $\log a$, below the rule-3/4 maximum
- Induced amplitude distribution not log-uniform: whatever the $\tilde{\delta}$ prior pushes forward (Fig. 4)

Samplers

- Poisson: $N \sim \text{Poisson}(\bar{n})$ uniform points
- Thomas / nested: top-level parents on W expanded by $5\times$ the total displacement s.d. (missed mass $e^{-12.5}$), offspring, crop to W
- Matérn II: primaries on $W \oplus R$ with $U(0, 1)$ marks, delete a point if a neighbour within R has a smaller mark, crop (exact)
- LGCP: circulant embedding on the $(2M)^2$ torus, spacing $1/M$: $Y = \mathcal{F}^{-1}[\sqrt{\lambda} \mathcal{F}w]$, w white noise, $\lambda = \mathcal{F}(\text{covariance})$ (exact on the grid); Poisson cell counts, uniform in cells
 - M : smallest power of two ≥ 128 with $|K_{\text{grid}} - K| \leq 0.1 \cdot 2\pi r s_0(r; \bar{n})$ on $R(\bar{n})$
 - Grid K : bilinear interpolation of e^C on the lattice, midpoint rule at $1/(4M)$

- Patterns conditioned on $n \in [20, 2000]$ (tables' domain), redrawn from the same stream

2 Implemented pipeline

Code

<code>src/cloudforger/simulation/seeding.py</code>	addressed random streams
<code>src/cloudforger/simulation/families.py</code>	rules, shape boxes, $K - \pi r^2$, CV, $\tilde{\delta}$, amplitude solve
<code>src/cloudforger/simulation/processes.py</code>	the five samplers
<code>src/cloudforger/simulation/lgcp_grid.py</code>	LGCP grid size
<code>src/cloudforger/simulation/sweep.py</code>	θ draw, replicate patterns, shards
<code>src/cloudforger/simulation/check.py</code>	mean $\hat{K}$ vs closed form
<code>scripts/simulate.py</code>	CLI: run , merge , check
<code>scripts/feasibility.py</code>	maximum reachable $\tilde{\delta}$
<code>scripts/simulation_figures.py</code>	figures of this document
<code>configs/simulation/config.yaml</code>	$\bar{n}$, $\tilde{\delta}$ range, rules, sizes

Configuration

- 10 000 θ per family $\times$ 2 patterns = 20 000 patterns per family, 100 000 total
- $\bar{n} \in [100, 800]$, $\tilde{\delta} \in [0.1, 4]$, $\varepsilon = 0.2$, $c = 0.45$, fill 0.95, $\mu_1 \geq 1$; shards of 250 θ

Seeding

- Key (DV = 3, set, family, index, role), root shared with 01
- θ : (sweep, family, i , PARAMS); pattern: (sweep, family, $2i + \text{rep}$, PATTERN)
- Checks: (pilot, family, $10^9 + 10^4 i + p$); gallery (pilot, $2 \cdot 10^9 + \dots$); both disjoint from the null curves of 01
- More θ later: append indices, nothing existing changes

Steps

1. **Run** (`run --jobs 10`; 3 min 10s on 10 cores): one file per shard, atomic, resumable
2. **Merge** (`merge`): per family `points.npz` (points, offsets) + `manifest.csv` in `data/simulation/`; ~ 106 MB each
 - Manifest: `case_id` (family-theta-rep), $\bar{n}$, $\tilde{\delta}$, CV, amplitude, shape, model parameters, LGCP M , n , tries
3. **Check** (`check`): 8 θ per family, 500 patterns each; $\tilde{K} = \hat{K} n(n-1)/\bar{n}^2$ (unbiased) vs closed form on 16 radii of $R(\bar{n})$

Results

Check	Result	Reference
Closed-form K vs old DV3 implementation	$\leq 10^{-10}$	—
Sampler $\tilde{K}$ vs K , max $ z $	1.6–2.8 by family	$\lesssim 3.4$ (16 radii)
Sampler $\tilde{K}$ vs K , max rel. error	$\leq 3.2\%$; Matérn 10% at $K \approx 0$	—
$\log \tilde{\delta}$ uniform, KS distance	≤ 0.0075	0.0136 (1%)
Shape draws per θ	1; nested mean 1.96, max 14	—
Pattern redraws ($n \notin [20, 2000]$)	0; realised n 44–961	—
Realised max CV	Thomas 0.32, nested 0.30, LGCP 0.34	$c = 0.45$
LGCP grid M (128/256/512/1k/2k/4k)	8064/1013/566/268/76/13 θ	—
5% L-test size, Poisson pool	0.049 (20 000 patterns)	0.05

3 Regimes

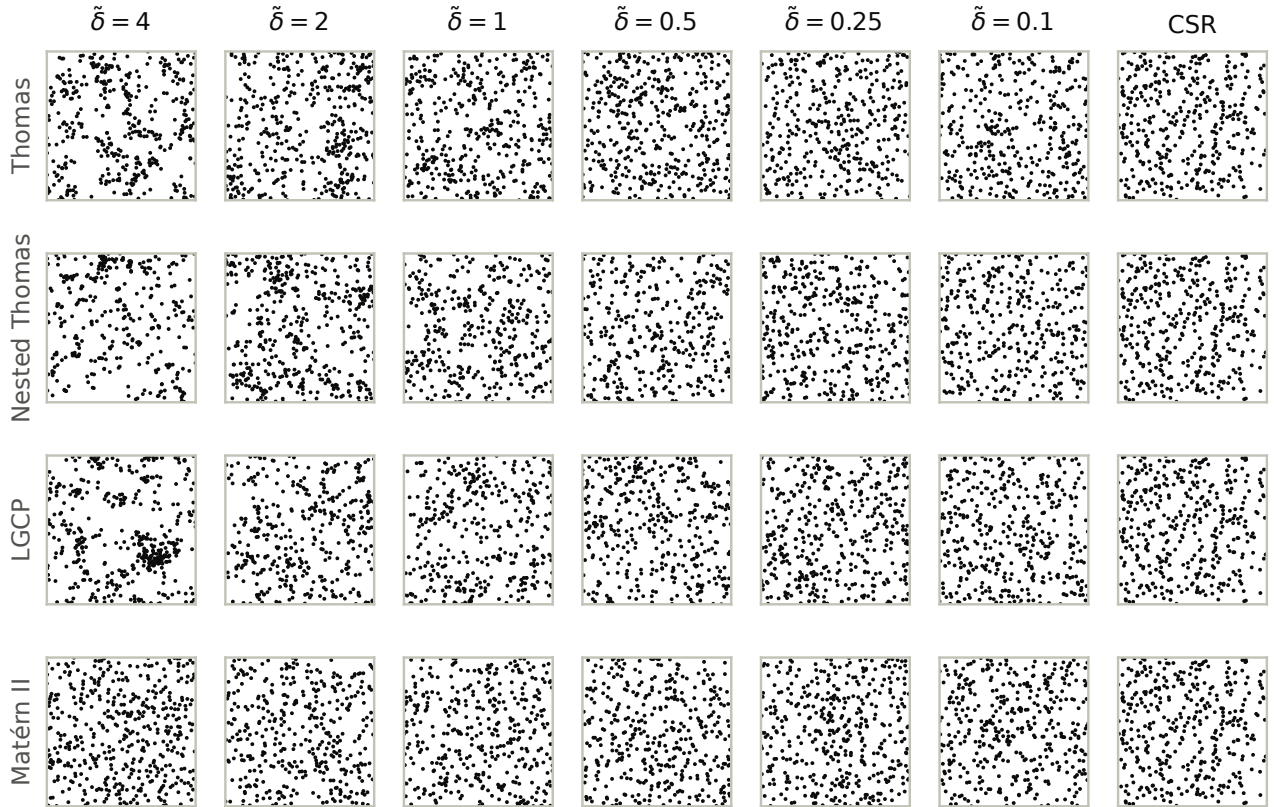


Figure 1: One pattern per cell, $\bar{n} = 400$, fixed shape per family (Thomas $\sigma = 0.03$; nested $\sigma_2 = 0.008$, $\rho = 5$, $\mu_1 = 3$; LGCP $s = 0.04$). Only the amplitude changes along a row.

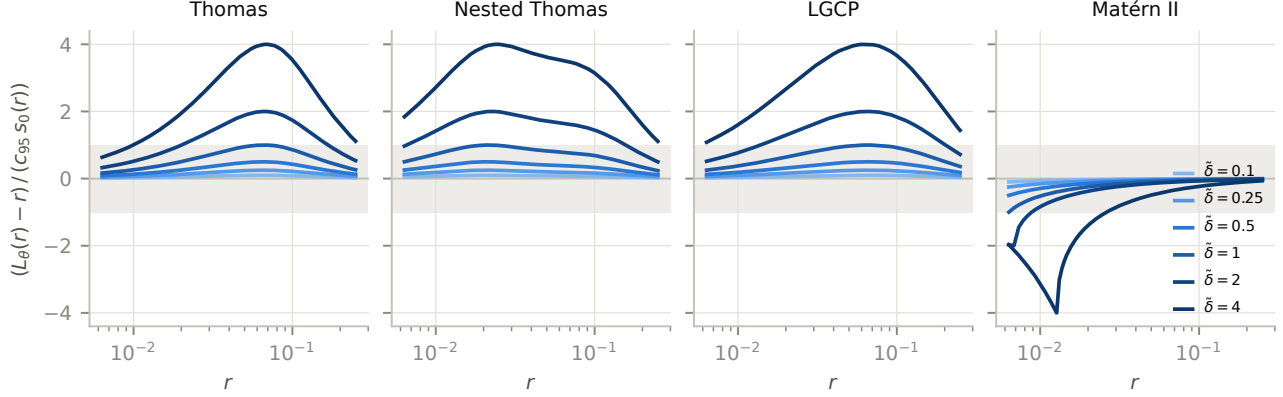


Figure 2: Closed-form $L_\theta(r) - r$ in units of the critical envelope, same θ as Fig. 1, on $R(400)$. Grey band: rejection boundary. Peak height = $\tilde{\delta}$.

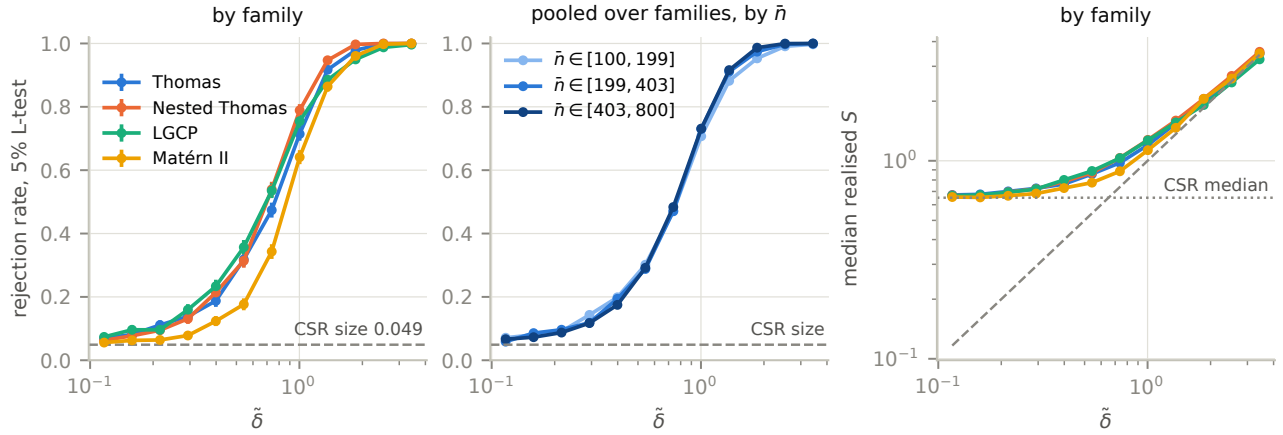


Figure 3: The 5% L-test of 01 on all 100 000 simulated patterns. Left: rejection rate by family, 12 log-bins of $\tilde{\delta}$, 95% intervals. Middle: pooled over families, by $\bar{n}$ tercile. Right: median realised S (dashed: $S = \tilde{\delta}$; dotted: CSR median).

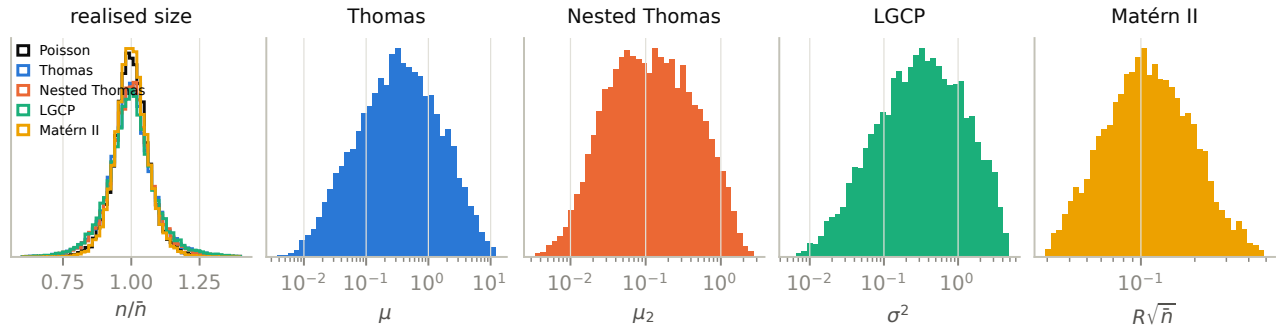


Figure 4: Left: realised $n/\bar{n}$. Right: induced distribution of the solved amplitude (Matérn: $\tau = R\sqrt{\bar{n}}$).

What the figures show

- **Convergence to Poisson** (Figs. 1, 2): along each row only the amplitude moves; curves shrink uniformly into the null band; at $\tilde{\delta} \leq 0.25$ patterns are visually CSR
- **Scale structure survives the amplitude:** nested keeps its two-step shoulder at every $\tilde{\delta}$; Matérn's deficit sits just above $r_{\min}(\bar{n})$
- **Same regime, same difficulty across families** (Fig. 3, left):

$\tilde{\delta}$ bin	(0.1, 0.2]	(0.2, 0.4]	(0.4, 0.8]	(0.8, 1.25]	(1.25, 2]	(2, 4]
Thomas	0.078	0.136	0.339	0.712	0.952	0.999
Nested	0.072	0.135	0.366	0.779	0.974	1.000
LGCP	0.085	0.152	0.386	0.746	0.923	0.988
Matérn II	0.059	0.084	0.214	0.639	0.914	0.997

- $\tilde{\delta}$ **absorbs** $\bar{n}$ (middle): terciles [100, 199], [199, 403], [403, 800] agree within 0.03 in every bin
- **Floor and ceiling:** below $\tilde{\delta} \approx 0.25$ rejection ≤ 0.15 and median $S = \text{CSR median}$ (0.65); above $\tilde{\delta} \approx 1.25$ rejection ≥ 0.91 and median $S \approx \tilde{\delta}$
- **Correction to 01:** at $\tilde{\delta} \approx 1$ rejection is 0.64–0.78, not $\approx 50\%$: pattern noise adds to the sup, so S tends to exceed $\tilde{\delta}$
- **Replicates:** the two patterns of a θ agree on rejection 77–82% of the time; correlation of $\log S$ 0.71–0.82

Coverage

$\tilde{\delta}$ **coverage.** Cells: 6 $\log\tilde{\delta}$ bins $\times$ 3 $\log\bar{n}$ bins, expected 556 θ each.

Family	realised $\tilde{\delta}$	$\log\tilde{\delta}$ uniform (KS p)	$\bar{n} \perp \tilde{\delta}$ (χ^2 p)	θ per cell
Thomas	0.1001–3.9998	0.76	0.81	530–581
Nested	0.1000–3.9969	0.98	0.54	519–587
LGCP	0.1000–3.9984	0.78	0.44	524–589
Matérn II	0.1001–3.9939	0.60	0.43	514–601

Parameter coverage at each $\tilde{\delta}$. At fixed $(\bar{n}, \tilde{\delta})$ the allowed parameters are the shapes passing rules 1–2 at $\bar{n}$; the amplitude is then determined.

- **By construction:** shape drawn after $\tilde{\delta}$, from its own uniforms; replay of every θ with $\tilde{\delta} < 0.2$ or $\tilde{\delta} > 3$: 0 shapes rejected by the amplitude solve (only nested geometric rejections) $\Rightarrow$ at every $\tilde{\delta} \leq 4$ the shape distribution is the full rule-1/2 prior, independent of $\tilde{\delta}$
- **Range:** normalised position $u = \log(x/x_{lo})/\log(x_{hi}/x_{lo})$ in the box at $\bar{n}$ spans 0.000–1.000 in every $\tilde{\delta}$ bin (Thomas σ , LGCP s , nested μ_1); nested σ_2, ρ reach 0.97–0.99 (top corner of the (σ_2, ρ) region excluded by the outer-scale rule)
- **Uniformity:** Thomas, LGCP uniform in u within each $\tilde{\delta}$ bin (KS p 0.08–0.94); u -quintile $\perp \tilde{\delta}$ bin (χ^2 p 0.10–0.81, all shape coordinates)
- **Nested:** vs a direct rejection sample of the allowed region at the same $\bar{n}$: pooled KS p 0.53–0.93; 3 of 18 per-bin tests < 0.05 (multiple testing; excluded as bias by the replay)
- **Amplitude range at fixed $\tilde{\delta}$** (realised within $\pm 10\%$ vs grid over $\bar{n}$ and allowed shape):

$\tilde{\delta}$	Thomas μ		LGCP σ^2		Matérn R	
	realised	theory	realised	theory	realised	theory
0.1	0.0037–0.32	0.0033–0.35	0.0055–0.28	0.0049–0.27	0.0010–0.0082	0.0010–0.0079
0.5	0.019–1.5	0.018–1.7	0.027–1.2	0.025–1.2	0.0022–0.017	0.0022–0.017
1	0.042–3.6	0.040–3.5	0.047–2.1	0.049–2.1	0.0029–0.022	0.0029–0.022
4	0.31–12	0.25–14	0.22–4.8	0.19–5.0	0.0059–0.047	0.0068–0.051

Theory grid coarse, so realised can exceed it

- **Density:** one-shape families $\geq 297 \theta$ per ($\tilde{\delta}$ bin, shape quintile); nested ≥ 62 per ($\tilde{\delta}$ bin, marginal quintile), but jointly $\approx 10^4 / (6 \cdot 5^3) \approx 13$ per cell

4 Statistical notes

Interpretation

- Regimes defined by one classical yardstick (L); visually, Matérn inhibition at $\tilde{\delta} \leq 4$ is hard to see by eye although the test sees it
 - $\bar{n} = 400$, $\tilde{\delta} = 4$: $R = 0.0127$ (0.25 spacings), $\pi R^2 \bar{n} = 0.20$; 25 fresh patterns all visually near-CSR, all with $S \in [3.76, 4.48]$; minimum pair distance $\geq R$ (sampler correct)
 - Removes only pairs closer than a quarter spacing; visibly regular patterns need fill ≈ 0.95 , i.e. $\tilde{\delta} \approx 8.6$ at $\bar{n} = 400$ (outside the common range)
- Matérn lags in $(0.2, 1.25]$: its departure is concentrated at small r , where $\hat{L}$ is biased and pairs are few; the centred test loses part of it (likely; not verified)
- Shape distribution at every $\tilde{\delta}$ is the full rule-1/2 prior (Coverage): no shape rejected by the amplitude solve; nested redraws are geometric (the ρ - σ_2 coupling)
- Allowed shape box depends on $\bar{n}$ (lower bound $\propto r_{\min}(\bar{n})$): in absolute units the smallest scales occur only at large $\bar{n}$; in normalised u , shape $\perp \bar{n}$
- Matérn II has no shape: $(\bar{n}, \tilde{\delta})$ determines $R \Rightarrow$ regime-conditional inference error for Matérn conditions on almost the whole parameter (within a $\tilde{\delta}$ bin R still varies $\approx 8\times$ via $\bar{n}$ and bin width); not comparable one-to-one with families that keep shape variation
- Rule 3 inactive in this range (max realised CV $0.34 < 0.45$): it matters only if $\tilde{\delta}$ is ever raised
- Induced amplitude marginals (Fig. 4) are the effective inference prior: learned estimators shrink toward them

Scope and limits

- $\tilde{\delta}$ at $\bar{n}$; realised $n/\bar{n}$ 5–95%: 0.89–1.11 (Poisson, Matérn), 0.85–1.15 (clustered)
- Gallery and curves at one shape per family and $\bar{n} = 400$: illustrative, not exhaustive; the power table uses the whole pool
- LGCP: grid process, bias bounded at $0.1 s_0$; exact circulant embedding on a $2\times$ torus
- Thomas/nested buffer truncation $e^{-12.5}$

Open

- δ_{lo} : the classical test is at its floor below ≈ 0.25 ; decide with learned methods' curves
- Carving train/validation/test from the pool (by θ , both replicates together)
- Nested joint shape coverage thin ($\approx 13 \theta$ per $\tilde{\delta}$ bin $\times 5^3$ cell): enough for averaging over shape, not for joint shape analysis; more θ can be appended
- Shard files (`data/simulation/shards/`, 626 MB) redundant after merge